Problem: Hotel workforce planning with an adjustable room count

Everything in this problem is driven by one number, **R = number of rooms**. Change R at the top of the notebook and every formula, table and plot updates. The same code works for a 10-room guesthouse or a 300-room resort.

1. The scenario

A hotel with **R rooms** and a small restaurant wants to answer four questions for the year:

1. How many regular housekeepers and how many overtime hours give the lowest daily cost? (**Differentiation**)
2. How many waiters are needed at each hour, and is a fixed roster efficient? (**Integration**)
3. How long do staff stay, and what does turnover cost? (**Sequences and series**)
4. In which months is temporary staff needed? (**Fourier and DCT**)

2. Parameters that scale with R

Parameter	**Formula**	**R = 15**	**R = 60**	**R = 150**
Linear cost coefficient a	2R/3	10	40	100
Linear cost coefficient b	2R/5	6	24	60
Fixed daily overhead c	$8R^2/45$	40	640	4,000
Peak waiters P	about R/4	4	15	38
Base waiters w_0 (at open and close)	about P/4	1	4	10
Fixed roster n	about 0.8 × P	3	12	30

The overhead grows with R^2 so the minimum cost stays positive at every size. You can change any of these. The formulas in the next sections work for whatever values you choose, which is what makes it a general model.

3. Section 1: Differentiation (housekeepers and overtime)

Model (thousand LKR per day), with x = regular housekeepers and y = overtime hours:

$C(x, y) = 2x^2 + 3y^2 - 2xy - a \cdot x - b \cdot y + c$

Why this shape? The $2x^2$ and $3y^2$ terms mean cost rises faster as you add more of either. The $-2xy$ term means regular staff and overtime partly substitute for each other. The negative linear terms mean a little of each is cheaper than none, because unserviced rooms cost money. Together these create a bowl shape with a lowest point.

Steps:

1. **Partial derivatives:** $C_x = 4x - 2y - a$ and $C_y = 6y - 2x - b$.
2. **Critical point:** set both to 0. This is two equations in two unknowns, and solving them gives general formulas:
    - $x^* = (3a + b)/10$, which equals **0.24R**
    - $y^* = (a + 2b)/10$, which equals **about 0.147R**
3. **Second derivatives:** $C_{xx} = 4$, $C_{yy} = 6$, $C_{xy} = -2$.
4. **Discriminant:** $D = C_{xx} \cdot C_{yy} - (C_{xy})^2 = 24 - 4 =$ **20**.
5. **Classify:** $D > 0$ and $C_{xx} > 0$, so it is a **minimum**, for every R. D doesn't depend on R, so the classification never changes.
6. **Minimum cost:** C_min = $c - (a \cdot x^* + b \cdot y^*)/2$, which equals **about 0.0684R^2**.
7. **Round to whole staff** and compute the cost penalty.

R	x*, y*	Rounded (x, y)	C_min	C at rounded	Penalty
15	3.6, 2.2	4, 2	15.4	16.0	0.6
60	14.4, 8.8	14, 9	246.4	247.0	0.6
150	36, 22	36, 22	1,540	1,540	0

Plots: a 3D surface (plot_surface) and a contour plot with the optimum marked. On the contour plot, the ellipses shrink toward the minimum.

Viva questions:

- Why does $D > 0$ with $C_{xx} > 0$ mean a minimum? What would $D < 0$ mean (a saddle point)?
- Why can't you just set overtime to 0?

4. Section 2A: Integration (waiter demand across the day)

Model: waiters needed t hours after opening ($0 \le t \le 12$), a parabola that starts at w_0, peaks at P at t = 6, and returns to w_0:

$w(t) = w_0 + A \cdot t(12 - t)/36$, where $A = P - w_0$

For R = 15 ($w_0 = 1$, P = 4) this is $1 + t - t^2/12$.

General results:

- **Peak:** $w'(t) = 0$ at t = 6, and $w'' < 0$ confirms a maximum.
- **Total staff-hours (area under the curve):** $\int_0^{12} w(t)\,dt$ = **$12w_0 + 8A$**.
- **Daily wage cost:** total × 450 LKR.
- **Times when a fixed roster of n is not enough:** solve $w(t) = n$, which gives $t = 6 \pm h$ with $h = 6\sqrt{((P - n)/A)}$.
- **Understaffed staff-hours:** $\int(w - n)\,dt$ between those two times = **$(4/3)\cdot h\cdot(P - n)$**.
- **Idle staff-hours:** the area where n exceeds w(t), which equals **12n − total + understaffed**.

R	Total staff-hours	Daily wage cost	Roster n	Not enough between	Understaffed	Idle
15	36	LKR 16,200	3	$t \approx 2.54$ to 9.46	4.62	4.62
60	136	LKR 61,200	12	$t \approx 2.87$ to 9.13	12.53	20.53
150	344	LKR 154,800	30	$t \approx 2.79$ to 9.21	34.21	50.21

Self-check: total − understaffed + idle should equal n × 12. For R = 60: 136 − 12.53 + 20.53 = 144 = 12 × 12 ✓.

Three ways to verify each integral (this is where the marks for turning maths into code come from):

1. **SymPy** for the exact answer.
2. **scipy.integrate.quad** for a numerical answer.
3. **A trapezoid rule** you code yourself, and show the error shrinking as you increase the number of strips.

5. Section 2B: Sequences and series (staff turnover)

Idea: if a staff member stays each year with probability r, expected years of service form a geometric series:

$1 + r + r^2 + r^3 + \ldots = 1/(1 - r)$, which converges only when $|r| < 1$.

Department	r	Expected tenure (sum to infinity)	Terms for 99% ($r^n \leq 0.01$)
Kitchen	0.70	3.33 years	13
Front desk	0.80	5 years	21
Housekeeping	0.90	10 years	44

Linking it to R: the number of housekeepers is x*, and hires needed per year ≈ x* × (1 − r). For example, with R = 60 and r = 0.8, that is about 14.4 × 0.2 ≈ 3 hires a year. Multiply by a cost per hire (choose your own figure) to get annual turnover cost.

Plots: partial sums S_n climbing toward the limit, with a horizontal line at 1/(1 − r) for each r. Be ready to explain why the series would diverge for r ≥ 1.

6. Section 3: Transforms (seasonality)

Data (monthly occupancy %): 78, 82, 70, 65, 52, 48, 60, 68, 55, 58, 66, 90. The average is 66%. Occupancy is a percentage, so this data stays the same for any R. **Rooms sold per night = R × occupancy/100** (for example, December at R = 60 is 54 rooms).

Steps:

1. **DFT:** numpy.fft.fft on the 12 values. The zero-frequency term is the sum (792). Find which non-zero frequency has the largest magnitude and explain what a cycle means in months.
2. **Fourier series:** map month m to x = 2πm/12. Compute a_0, a_n, b_n with the formula-sheet integrals (as sums or with quad), then plot 1, 2 and 3 harmonics over the data. Comment on how the fit improves.
3. **DCT compression:** scipy.fft.dct(x, norm='ortho'), keep the largest k coefficients, rebuild with idct, and plot error against compression ratio 12/k.
4. **Staffing decision using R:** scale staff by each month's occupancy relative to the average:
    - Housekeepers in month m = x* × occ_m/66
    - Peak waiters in month m = P × occ_m/66

Months where this exceeds your regular roster need temporary staff. This connects Section 3 to Sections 1 and 2, which makes the case study one coherent story. Use the smoothed curve from step 2 or 3 for the decision.

7. The parameter cell at the top of the notebook

```
R = 60                  # rooms, change this only
a, b, c = 2*R/3, 2*R/5, 8*R**2/45
P = round(R/4); w0 = round(P/4); n = round(0.8*P)
wage = 450              # LKR per staff-hour
occ = [78,82,70,65,52,48,60,68,55,58,66,90]
```

Everything else refers to these variables, so changing R reruns the whole analysis.

8. Notebook structure (check names against the rubric)

1. **Differentiation:** the cost model, partial derivatives, discriminant, plots, rounding.
2. **Integration and series:** the waiter curve, three verification methods, turnover series.
3. **Transforms:** DFT, Fourier series, DCT, the temporary-staff decision.